Mass Gap for the O(N) Model at Large N via Brascamp-Lieb and Feynman-Kac

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Abstract

We outline a proof of the mass gap for the O(N) model in two dimensions at large N. The σ-field effective action $s_{\text{eff}}[\sigma]$ has a minimum at σ_* determined by the gap equation, which gives an exponentially small mass $m_0 = \sqrt{\sigma_*}$ by dimensional transmutation. When the Hessian of s_{eff} at σ_* is positive, Brascamp-Lieb gives σ-concentration with $1/N$ variance, and the Feynman-Kac representation controls the $1/N$ corrections to the propagator mass.

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1 Setup

1.1 Model

O(N) LSM on $\Lambda \subset \mathbb{Z}^2$:

$$S[\varphi] = \tfrac{1}{2}\sum_i \langle \varphi^i, (-\Delta)\varphi^i\rangle + N\sum_x \lambda\big(\tfrac{|\varphi(x)|^2}{N} - v^2\big)^2, \qquad v^2 = R^2/N. \tag{1}$$

1.2 Hubbard-Stratonovich

With $\sigma(x) = |\varphi(x)|^2/N$, integrating out φ:

$$\langle \varphi^i(x)\varphi^i(0)\rangle = \mathbb{E}_\sigma\big[(-\Delta+\sigma)^{-1}(x,0)\big], \tag{2}$$

where $\mu_\sigma \propto e^{-N s_{\text{eff}}[\sigma]}$ with

$$s_{\text{eff}}[\sigma] = \tfrac{1}{2}\mathrm{Tr}\log(-\Delta+\sigma) + \sum_x \lambda(\sigma(x) - v^2)^2. \tag{3}$$

2 The gap equation and the saddle point σ_*

2.1 The minimum of s_{eff}

For spatially uniform $\sigma(x) = \sigma_0$, the derivative is

$$\frac{ds_{\text{eff}}}{d\sigma_0} = \tfrac{1}{2}G(x,x;\sigma_0) + 2\lambda(\sigma_0 - v^2), \tag{4}$$

where $G(x,x;\sigma_0) = \int \frac{d^2k}{(2\pi)^2}\frac{1}{k^2+\sigma_0}$ is the lattice Green's function diagonal. Since $G > 0$, the function $s_{\text{eff}}(\sigma_0)$ is *not* minimized at v^2 — the Tr log term pushes the minimum to $\sigma_* < v^2$.

The minimum σ_* satisfies the **gap equation**:

$$\boxed{\tfrac{1}{2}G(x,x;\sigma_*) = 2\lambda(v^2 - \sigma_*)} \tag{5}$$

In 2d with UV cutoff Λ: $G(x,x;\sigma_*) \approx \frac{1}{4\pi}\log\frac{\Lambda^2}{\sigma_*}$, giving

$$\frac{1}{4\pi}\log\frac{\Lambda^2}{\sigma_*} = 4\lambda(v^2 - \sigma_*). \tag{6}$$

For $\sigma_* \ll v^2$: $\sigma_* = \Lambda^2 e^{-16\pi\lambda v^2}$, which is exponentially small — this is dimensional transmutation.

2.2 The physical mass

The φ-propagator mass at leading order ($N = \infty$) is

$$m_0 = \sqrt{\sigma_*}, \tag{7}$$

since $\sigma = \sigma_*$ is deterministic at $N = \infty$ and $(-\Delta + \sigma_*)^{-1}(x,0) \sim e^{-m_0|x|}$.

3 Brascamp-Lieb concentration at σ_*

3.1 The Hessian at σ_*

The Hessian of s_{eff} at σ_* in Fourier:

$$H(k) = 2\lambda - \tfrac{1}{2}\Pi(k), \qquad \Pi(k) = \int \frac{d^2p}{(2\pi)^2} \frac{1}{(p^2+\sigma_*)((k-p)^2+\sigma_*)}. \tag{8}$$

At $k = 0$: $\Pi(0) = 1/(4\pi\sigma_*)$, so

$$H(0) = 2\lambda - \frac{1}{8\pi\sigma_*}. \tag{9}$$

3.2 Convexity condition

The Hessian is positive definite when $H(0) > 0$, i.e.,

$$\boxed{\sigma_* > \frac{1}{16\pi\lambda}} \tag{10}$$

This holds when the lattice is coarse enough that σ_* (from the gap equation) hasn't yet become exponentially small. Specifically, from (**??**): $\sigma_* > 1/(16\pi\lambda)$ iff $\log(\Lambda^2 \cdot 16\pi\lambda) < 16\pi\lambda v^2$, i.e., $\Lambda^2 < e^{16\pi\lambda v^2}/(16\pi\lambda)$.

When (**??**) holds, the convexity parameter is $\kappa = 2\lambda - 1/(8\pi\sigma_*) > 0$ and Brascamp-Lieb gives:

$$\mathrm{Var}_{\mu_\sigma}(\sigma(x)) \le \frac{1}{\kappa N}, \qquad |\kappa_k(\sigma(x))| \le \frac{C^k\, k!}{(\kappa N)^{k-1}}. \tag{11}$$

3.3 When convexity fails

For $\Lambda^2 > e^{16\pi\lambda v^2}/(16\pi\lambda)$ (fine lattice or continuum limit): σ_* becomes too small, the Hessian goes negative at $k = 0$, and Brascamp-Lieb does not apply at σ_*.

This is the regime where a multiscale or cluster expansion argument is needed (as in Kupiainen [2]). The BL approach is limited to lattices that are coarse enough for convexity.

4 Feynman-Kac and $1/N$ corrections

When the convexity condition (**??**) holds:

4.1 Leading order

At $N = \infty$: $\sigma = \sigma_*$ deterministically, mass $= m_0 = \sqrt{\sigma_*}$. This requires no further analysis — just the gap equation.

4.2 $1/N$ corrections via Feynman-Kac

At finite N: σ fluctuates around σ_* with $\mathrm{Var} = O(1/(\kappa N))$. The FK cumulant expansion:

$$\mathbb{E}_\sigma[e^{-\int \sigma(\omega)\,dt}] = \exp\bigl(\kappa_1 + \tfrac{1}{2}\kappa_2 + \tfrac{1}{6}\kappa_3 + \cdots\bigr) \tag{12}$$

with $\kappa_1 = -s\sigma_*$ (leading mass) and $\kappa_k = O(s/(\kappa N)^{k-1})$ for $k \geq 2$ (from BL bounds).

The corrected mass:

$$m^2 = \sigma_* - \frac{\delta\sigma}{N} + O(1/N^2) \tag{13}$$

where $\delta\sigma/N$ is computable from the σ-propagator H^{-1} integrated along Brownian paths. For $N \gg \delta\sigma/\sigma_*$: $m^2 > 0$ and the mass gap persists.

4.3 Mass gap theorem (convex regime)

Theorem 1. *For the O(N) LSM on $\mathbb{Z}^2$, if the convexity condition (**??**) holds, there exists $N_0 = O(\delta\sigma/\sigma_*)$ such that for $N \geq N_0$:*

$$|\langle \varphi^i(x)\varphi^i(0)\rangle_c| \leq C\, e^{-m|x|} \tag{14}$$

with $m^2 = \sigma_*(1 + O(1/N)) > 0$ and σ_* from the gap equation (12).

5 Beyond the convex regime

When (**??**) fails (fine lattice / continuum limit):

- σ_* is exponentially small, $H(0) < 0$, BL fails
- The σ-measure is not log-concave at the saddle point
- A multiscale analysis is needed to control the $1/N$ expansion scale-by-scale (each momentum shell is individually convex in the UV)
- This is the regime addressed by Kupiainen [2] via cluster expansion
- An RG approach on σ-invariants, using $1/N$ at each scale, may provide an alternative

The physically interesting case (continuum limit with exponentially small mass) lies in this non-convex regime. Extending the proof to cover it is the main open problem.

6 The $N = 2$ obstruction

For $N = 2$, the BKT transition occurs at finite λ (Dario-Garban [4]), driven by vortices ($\pi_1(S^1) = \mathbb{Z}$). The σ-concentration cannot detect angular topology. For $N \geq 3$: $\pi_1(S^{N-1}) = 0$, no BKT.

References

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